
Yonder: A 4.65M-Frame Drone Navigation Dataset and the Cross-Simulator Generalization Gap

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Vision-language drone navigation has advanced rapidly through the use of open-
2 vocabulary detectors fine-tuned on simulation data, but evaluation methodol-
3 ogy has not kept pace. We introduce **Yonder**, a 4.65-million-frame drone-
4 perspective dataset spanning 167 indoor environments (all with semantic anno-
5 tations) with stereo RGB, depth, LiDAR, semantic segmentation, and dense way-
6 point coverage—the largest publicly available training resource for drone indoor
7 navigation. We then use Yonder to investigate a question that the dataset itself
8 cannot answer: whether offline detection improvements predict closed-loop nav-
9 igation success. Across four fine-tuning attempts using two open-vocabulary de-
10 tector architectures (OWL-ViT v2 and Grounding DINO), we achieve a $9.7\times$
11 improvement in offline detection mAP ($4.8\% \rightarrow 46.7\%$) on Yonder’s held-out
12 evaluation split. Yet across 3,000+ closed-loop navigation trials in Isaac Sim,
13 none of these fine-tuned models exceed the zero-shot baseline. The root cause
14 is a cross-simulator domain gap: bounding box geometry calibrated to Yonder’s
15 source simulator (Habitat-Sim) produces systematic 3D localization errors in the
16 evaluation simulator (Isaac Sim), including negative-altitude goal predictions that
17 cause the drone to dive into the floor. With the domain gap diagnosed and patched,
18 navigation success rate converges at 23–25% across all detector configurations—
19 fine-tuned and zero-shot—revealing that detection quality, the apparent binding
20 constraint, is not actually limiting performance. We argue that this constitutes a
21 general failure mode in evaluating simulation-trained perception: offline metrics
22 on the training simulator’s domain do not predict closed-loop performance on a
23 different simulator’s domain, even when both are physics-based platforms target-
24 ing the same task. We release Yonder, all four fine-tuned model checkpoints, the
25 closed-loop benchmark, and full evaluation logs to enable rigorous study of cross-
26 simulator generalization in embodied AI.

27 1 Introduction

28 The release of large open-vocabulary detectors—Grounding DINO [1], OWL-ViT [2], and Florence-
29 2 [3]—has reshaped vision-language drone navigation. Recent systems pair these detectors with
30 classical depth estimation and motion planning to achieve impressive performance on operational
31 navigation tasks [4, 5]. The standard recipe is straightforward: collect drone-perspective imagery in
32 a simulation environment, fine-tune the detector on this domain, and deploy the fine-tuned model in
33 a downstream navigation pipeline.

34 This paper documents what happens when this recipe is applied at scale, with rigorous closed-loop
35 evaluation, and what the results imply for the field’s evaluation practices.

36 We make three contributions:

37 **1) Yonder dataset.** We construct a 4.65-million-frame drone-perspective dataset across 167 indoor
 38 environments (all from HSSD), with per-pixel semantic segmentation for all 167. Yonder includes
 39 stereo RGB, monocular depth, 360° LiDAR point clouds, semantic segmentation, and dense 1-meter
 40 waypoint coverage with 12 yaw headings per waypoint. Total dataset size is approximately 3.3 TB.
 41 To our knowledge this is the largest publicly available dataset for drone indoor navigation training,
 42 and the first to combine all five sensor modalities at this scale.

43 **2) Empirical demonstration that offline detection improvements do not predict closed-loop**
 44 **navigation success.** We fine-tune two open-vocabulary detector architectures using four different
 45 protocols on Yonder, achieving up to $9.7\times$ improvement in offline mAP@0.5. Across 3,000+ closed-
 46 loop navigation trials in Isaac Sim, none of these fine-tuned models exceed the zero-shot baseline.
 47 Detection quality, measured by standard offline metrics, has no relationship to closed-loop success
 48 in our experiments.

49 **3) Diagnosis of the cross-simulator domain gap as the underlying failure mode.** Through sys-
 50 tematic root-cause analysis, we identify that bounding box geometry calibrated to one simulator’s
 51 rendering conventions produces systematic 3D localization errors when transferred to another simu-
 52 lator. The failure manifests differently across detector architectures (confidence calibration collapse,
 53 negative-altitude goals, lateral localization shift), but the underlying cause is consistent: simulation-
 54 trained perception inherits simulator-specific geometric biases that are invisible to offline evaluation
 55 metrics.

56 We argue these findings have implications beyond drone navigation. Any system that trains percep-
 57 tion on one simulator and evaluates on another may be subject to similar gaps. The rigor of offline
 58 evaluation cannot substitute for closed-loop validation, and offline mAP improvements that do not
 59 transfer to deployment performance should be considered with skepticism rather than celebrated as
 60 progress.

61 2 Related Work

62 **Drone-perspective datasets.** Existing drone datasets focus primarily on outdoor settings and aerial
 63 imagery [6]. The few indoor drone datasets available are typically small (under 100K frames) and
 64 capture single environments. Yonder is approximately $50\times$ larger than the largest prior indoor drone
 65 dataset and spans more than $100\times$ as many environments.

66 **Open-vocabulary detection in robotics.** Grounding DINO [1], OWL-ViT [2], and similar detec-
 67 tors have become standard components in vision-language navigation pipelines [7, 8]. Fine-tuning
 68 these detectors on domain-specific data is widely reported to improve downstream task performance,
 69 though most reports rely on offline evaluation metrics.

70 **Simulation-to-real transfer.** The sim-to-real gap is well-studied [9, 10]. Domain randomization,
 71 domain adaptation, and simulation fidelity improvements have all been proposed. Less attention has
 72 been paid to the *sim-to-sim* gap—the failure of models trained on one simulator to transfer to another.
 73 We are aware of no prior work systematically characterizing this gap for vision-language drone
 74 navigation, despite cross-simulator evaluation being common practice (e.g., training in Habitat-Sim
 75 and evaluating in Gibson, Isaac Sim, or Unreal Engine [11, 12]).

76 **Closed-loop evaluation of navigation.** Several benchmarks evaluate navigation in closed loop [13,
 77 11]. These benchmarks generally use a single simulator for both training and evaluation, avoiding
 78 the cross-simulator scenario we study. Our work uses the closed-loop benchmark from [4], extended
 79 with statistical validation across multiple seeds.

80 3 The Yonder Dataset

81 3.1 Collection Methodology

82 Yonder was constructed by flying a simulated Holybro x500v2 quadcopter through 167 indoor 3D
 83 scenes from HSSD in Habitat-Sim [11], collecting multi-modal sensor data at dense waypoints. The
 84 drone is configured to match the physical Holybro x500v2: 500 mm wheelbase, 0.25 m safety radius,
 85 1.2 m cruising altitude.

86 **Waypoint sampling.** We use Habitat-Sim’s navmesh to sample navigable waypoints at 1.0 m base
 87 spacing. Adaptive densification adds waypoints at 0.1 m resolution near tight spaces (depth/IR <
 88 0.7 m) and at semantic frontiers (where new object IDs enter the camera frustum). The result is
 89 387,527 waypoints across the 167 scenes.

90 **Sensor configuration.** At each waypoint, the drone performs a 360° scan at 12 yaw headings
 91 (30° increments), capturing eight sensor streams per heading: stereo left RGB (480×640, 90°
 92 FOV, +15 cm forward, −5 cm lateral from COG); stereo right RGB (480×640, +15 cm forward,
 93 +5 cm lateral); forward monocular depth (480×640, co-located with stereo center); landing cam-
 94 era (240×320, +5 cm forward, downward-facing). Per-waypoint sensors (yaw-independent): up IR
 95 (64×64 depth array, +10 cm above COG), down IR (64×64, −15 cm below COG), 360° LiDAR
 96 point cloud (~230K points, synthesized from the 12 depth panoramas), position (XYZ in world
 97 coordinates).

98 **Total scale.** 4,650,324 frames across 167 scenes with 309 distinct semantic categories.

99 3.2 Scene Sources

100 Yonder is rendered from a single open-source 3D scene dataset, summarized in Table 1. An ear-
 101 lier collection pass also covered ReplicaCAD (84 scenes, CC-BY-4.0, no semantics), Replica (18
 102 scenes, Meta research-only terms), and HM3D / standalone scenes (6 scenes, Matterport academic-
 103 use EULA, per-user agreement required). The Replica and HM3D subsets are excluded from the
 104 public release because their upstream licenses do not permit open redistribution of derivative renders.
 105 The ReplicaCAD subset is permissively licensed but lacks semantic annotations and was not used
 106 in any reported experiment, so we drop it as well in service of a single-source, fully-experiment-
 107 relevant artifact.

Table 1: Yonder scene source (public release)

Source	Upstream license	Scenes	Waypoints	Has Semantics
HSSD	CC-BY-NC-4.0	167	387,527	167 of 167

108 For all 167 HSSD scenes, we re-rendered semantic segmentation channels (not stored in initial
 109 collection) using a separate pass through Habitat-Sim, then projected per-pixel semantic instance
 110 masks to 2D bounding boxes to produce COCO-format annotations: 24,001,088 bounding boxes
 111 across 4,032,975 annotated images and 309 categories.

112 3.3 Storage and Access

113 Yonder is hosted on the HuggingFace Hub at [https://huggingface.co/datasets/astralhf/](https://huggingface.co/datasets/astralhf/yonder)
 114 [yonder](https://huggingface.co/datasets/astralhf/yonder). Each waypoint is stored as a single compressed NPZ file (52 sensor arrays + optional
 115 semantic channels). Total storage: approximately 3.3 TB. We provide loading utilities, manifest files,
 116 and per-scene COCO annotation files in the accompanying code release. For reviewers and others
 117 wanting a fast preview before downloading the full release, we additionally publish a ~500 MB
 118 single-scene sample at <https://huggingface.co/datasets/astralhf/yonder-sample>.

119 3.4 Generation Cost

120 The released subset was generated for approximately \$60 in compute cost using rented RTX 4090
 121 GPU instances on Vast.ai. Generation took approximately 3 hours of wall-clock time using a fleet
 122 of 50 concurrent worker instances. This places Yonder-scale dataset generation within reach of
 123 small research groups and individual practitioners, in contrast to datasets that require dedicated
 124 infrastructure investments.

125 3.5 Intended Use

126 Yonder is intended for training drone-perspective perception models—open-vocabulary detectors,
 127 depth estimators, semantic segmentation, instance recognition, and similar tasks. All 167 scenes
 128 carry per-pixel semantic instance labels and support supervised training.

We explicitly note that **Yonder is not intended for end-to-end navigation policy training**. The dataset captures static viewpoint sensor data, not closed-loop trajectories with physics, collisions, and replanning. Models trained on Yonder must be evaluated in a closed-loop simulator (or on real hardware) before performance claims can be made.

A full Datasheet for Yonder, structured per [15], is provided in Appendix A.

4 Benchmark and Evaluation Protocol

We use the closed-loop drone navigation benchmark from [4], which evaluates an autonomous drone agent on natural language navigation commands in Isaac Sim with the Pegasus Simulator [12]. The benchmark covers three indoor environments (warehouse, hospital, office) with 51–67 navigation tasks per evaluation run, organized into 14 difficulty tiers ranging from stationary baselines through compound multi-step commands.

4.1 Closed-Loop Protocol

Each trial runs an IRIS quadrotor with a Mellinger controller in Isaac Sim. The drone receives a natural-language command (e.g., “fly to the forklift”), captures RGB-D observations at 0.5 Hz, runs the full perception-planning-control loop, and executes flight commands until success (within 5 m of the ground-truth target), failure (timeout at 90 s), or collision. We record final position error, collision-free flag, and detection metadata.

Metrics. Primary metric is success rate at 5 m (SR@5m): the fraction of trials where final position is within 5 m of ground truth. Secondary metrics include mean position error, collision-free rate (CF%), and detection success rate.

Statistical validation. We run 5 independent random seeds per condition (seeds 1000, 2000, 3000, 4000, 5000), with 93 trials per seed. Total: 465 closed-loop trials per condition. We report mean SR@5m with 95% bootstrap confidence intervals (10,000 resamples) and pairwise significance using permutation tests (10,000 permutations) with Bonferroni correction for multiple comparisons.

4.2 Detection Pipeline

The navigation pipeline (Track A from [4]) applies the *separation principle*: an open-vocabulary detector identifies the target object in the image, monocular depth estimation [14] computes its 3D position via backprojection through the camera intrinsics, and a classical motion planner with depth-conditioned safety executes the flight. The detector is the only component we modify across conditions.

5 Fine-Tuning Experiments

5.1 Setup

We fine-tune two open-vocabulary detector architectures using four protocols:

A1: OWL-ViT v2 fine-tuning. OWL-ViT v2-large fine-tuned on 29K frames from 90 Yonder scenes for 5 epochs. Detection heads only (frozen backbone). Pseudo-labels derived from semantic instance masks projected to bounding boxes.

A2: Grounding DINO Round 6. Grounding DINO-tiny fine-tuned on 40K frames from 20 Yonder scenes. Backbone frozen for first 3 epochs, then last 2 transformer blocks unfrozen. $lr_heads = 1e-5$, $lr_backbone = 5e-6$. Built-in HuggingFace bipartite matching loss with label smoothing (0.1). Calibration monitoring every 200 training steps.

A3: Grounding DINO Round 7 (cross-simulator mix). Same as A2, but training data mixes 90% Yonder Habitat-Sim frames (8,280 samples) with 10% Isaac Sim frames (920 samples) to test whether including target-domain data closes the gap. Pseudo-labels for Isaac Sim frames generated by zero-shot Grounding DINO with confidence threshold 0.20.

173 **A4: Grounding DINO Round 6 with Z-floor clamp.** Same checkpoint as A2, but with a runtime
 174 fix applied to the navigation pipeline: `goal_z = max(goal_z, 0.5)` to prevent sub-floor goal
 175 predictions.

176 5.2 Offline Detection Results

177 We measure mAP@0.5 and mAP@0.75 on Yonder’s held-out test split (17 scenes never seen during
 178 training, environment-level split, not frame-level). Results in Table 2.

Table 2: Offline detection performance on Yonder test set. N/M = not measurable due to confidence calibration collapse.

Model	mAP@0.5	mAP@0.75	Blank max score
OWL-ViT v2 zero-shot	4.8%	2.7%	0.42
OWL-ViT v2 fine-tuned (A1)	N/M	N/M	0.97
Grounding DINO zero-shot	~5%	—	0.39
Grounding DINO fine-tuned R6 (A2)	46.7%	26.9%	0.115
Grounding DINO fine-tuned R7 (A3)	45.4%	24.6%	0.115

179 The headline number is a $9.7\times$ improvement in mAP@0.5 (from 4.8% to 46.7%) for Grounding
 180 DINO fine-tuning. The OWL-ViT v2 fine-tune produces invalid offline metrics due to confidence
 181 calibration collapse: the fine-tuned model assigns scores >0.97 to blank gray images, making any
 182 threshold-based detection metric meaningless.

183 5.3 Closed-Loop Navigation Results

184 We evaluate each detector configuration in closed-loop Isaac Sim across 5 seeds. Results in Table 3.

Table 3: Closed-loop navigation success rate (5 seeds, 465 trials per condition).

Configuration	SR@5m	95% CI	p vs zero-shot
Hover (do-nothing baseline)	10.0%	[7.4, 13.0]	—
Zero-shot Grounding DINO	32.9%	[28.8, 37.2]	baseline
Fine-tuned OWL-ViT v2 (A1)	11.6%	[8.8, 14.4]	<0.001 (worse)
Fine-tuned GDINO R6 (A2)	23.5%	[19.0, 28.4]	0.003 (worse)
Fine-tuned GDINO R6+clamp (A4)	25.5%	[20.7, 30.5]	0.012 (worse)
Fine-tuned GDINO R7 (A3)	25.5%	[20.7, 30.5]	0.012 (worse)

185 **The headline result: no fine-tuned model exceeds the zero-shot baseline.** Despite a $9.7\times$ im-
 186 provement in offline mAP, the best fine-tuned model (Grounding DINO R6 with Z-clamp) underper-
 187 forms the zero-shot baseline by 7.4 percentage points ($p=0.012$). The OWL-ViT v2 fine-tune is the
 188 worst performer, scoring barely above hover (11.6% vs 10.0%, n.s.).

189 The relationship between offline mAP and closed-loop SR@5m is not just weak—it is anticorrelated
 190 within our fine-tuned configurations, and the model with the highest offline mAP performs worse in
 191 closed-loop than the model with no fine-tuning at all.

192 5.4 Convergence at the Detection Plateau

193 A 2×2 ablation (fine-tuned vs zero-shot Grounding DINO $\times$ safety profiles enabled vs disabled,
 194 153 trials per condition) reveals a striking pattern: SR@5m converges at 23–25% across all four
 195 configurations, shown in Table 4.

196 No pair differs significantly (all $p > 0.25$). The fine-tuned and zero-shot detectors are statistically
 197 indistinguishable in closed-loop, despite the fine-tuned detector having $9.7\times$ the offline mAP. This
 198 convergence is the empirical signature that detection quality is not the binding constraint for naviga-
 199 tion success in our setup.

Table 4: 2×2 ablation: SR@5m (CF% in parentheses).

	Safety enabled	Safety disabled
Fine-tuned Grounding DINO	23.5% (86.9%)	20.4% (71.8%)
Zero-shot Grounding DINO	23.5% (86.9%)	24.7% (80.0%)

6 Root Cause: The Cross-Simulator Domain Gap

The four fine-tuning attempts fail in different ways, but a single underlying cause connects them: **fine-tuning on one simulator’s renders calibrates the detector to that simulator’s specific geometric and photometric conventions, which do not transfer to a different simulator.**

6.1 Failure Mode 1: Confidence Calibration Collapse (OWL-ViT v2)

OWL-ViT v2 fine-tuning produces a model that assigns confidence scores >0.97 to blank gray images. The hard 0/1 sigmoid targets used during fine-tuning push the detection head’s logits to extreme values, and the resulting calibration is destroyed. Threshold-based detection becomes meaningless: any threshold below the inflation floor sees all bounding box proposals as positive; any threshold above sees nothing.

This failure is detectable by inspecting score distributions on out-of-distribution inputs (blank images, random noise). It is invisible to standard offline mAP metrics because mAP is computed on examples where targets are present and reasonable bounding boxes can be matched.

6.2 Failure Mode 2: Negative-Altitude Goal Predictions (Grounding DINO Round 6)

Grounding DINO fine-tuning preserves confidence calibration (blank-image max score: 0.115) but causes a different failure: negative-altitude goal predictions in Isaac Sim warehouse trials.

The pipeline computes 3D goal position from a detection’s bounding box center and the corresponding depth value:

$$\text{goal}_z = z_{\text{spawn}} + d_{(c_y, c_x)} \cdot \cos(\theta_{\text{pitch}}) - \delta_{\text{camera}} \quad (1)$$

Floor-level objects (forklifts, pallet jacks) appear lower in the image frame in Habitat-Sim’s rendering than in Isaac Sim’s. Fine-tuning shifts the detector’s bounding box centers downward to match Habitat-Sim’s geometry. When applied to Isaac Sim frames, these lower bounding box centers, combined with Isaac Sim’s different depth scale, produce $\text{goal}_z < 0$ —goals below the floor.

We measured negative-Z goal frequency in 153 trials. Fine-tuned Grounding DINO produces negative-Z goals in 24% of warehouse trials, compared to 4% for zero-shot. The drone, instructed to fly to a sub-floor coordinate, descends from 1.0 m altitude to 0.13 m in approximately 8 steps and accumulates 35–45 collisions per trial as it oscillates near the floor. Warehouse collision-free rate drops from 89% [5] to 61%.

A simple Z-floor clamp $\text{goal}_z = \max(\text{goal}_z, 0.5)$ eliminates the floor-diving behavior and recovers warehouse CF% to 98%, but does not improve SR@5m: navigation success converges at 25.5% with or without the clamp.

6.3 Failure Mode 3: Pseudo-Label Collapse (Grounding DINO Round 7)

To address the cross-simulator gap, we mixed 10% Isaac Sim frames into the training data for Round 7. Since we had no human-annotated bounding boxes in Isaac Sim, we used pseudo-labels generated by zero-shot Grounding DINO itself.

This created a circular training signal. The Round 7 model learned to reproduce zero-shot Grounding DINO’s outputs on Isaac Sim frames, biased toward Habitat-Sim distribution by the 90% majority training data. Round 7 SR@5m: 25.5%—identical to Round 6 with Z-clamp. Warehouse CF% collapsed to 17.6%, indicating new collision modes from lateral bounding box shifts.

Pseudo-labels from a baseline detector cannot teach a fine-tuned detector to do better than the baseline. Closing the cross-simulator gap requires real ground-truth annotations in the deployment domain.

6.4 Why Offline Metrics Miss These Failures

In all three failure modes, offline mAP on Yonder’s test split looks excellent (45–47%). The metric is computed on Habitat-Sim frames against Habitat-Sim ground truth bounding boxes. The detector is, in fact, very good at finding objects in Habitat-Sim images.

The failure modes only manifest under three specific conditions absent from offline evaluation:

1. *Out-of-distribution inputs* (blank images, atypical scenes) reveal confidence calibration breakage. Standard mAP only evaluates positive examples.
2. *3D backprojection* translates 2D detection errors into 3D goal errors. Offline evaluation never executes this transformation.
3. *Closed-loop dynamics* compound small errors over time. A drone instructed to fly to a sub-floor coordinate accumulates collisions over many timesteps; a single-frame evaluation never sees this consequence.

These conditions are precisely what closed-loop evaluation captures and offline evaluation cannot. The $9.7\times$ offline mAP improvement is real and reproducible, but it does not correspond to improvement in any of the dimensions that matter for closed-loop deployment.

7 Implications for Evaluation Practice

Our results suggest several implications for how the field evaluates simulation-trained perception:

Closed-loop evaluation is not optional for navigation systems. Offline detection metrics on the training simulator’s distribution can improve dramatically without any improvement in deployment performance. Researchers reporting fine-tuning improvements should report closed-loop results in the deployment environment, not just held-out detection metrics.

Multi-seed statistical validation matters. A single seed of our Round 6 fine-tune produced $\text{SR}@5\text{m} = 48.4\%$, suggesting the fine-tune was a major improvement. The 5-seed result is 25.5%, statistically indistinguishable from baseline. Per-seed variance is large (4–20% range) and headline single-seed numbers can be misleading by $2\times$ or more.

Cross-simulator transfer should be tested explicitly. The community has tacitly assumed that physics-based simulators are interchangeable for perception training. Our results show this assumption can be wrong even between two well-established simulators (Habitat-Sim and Isaac Sim) targeting the same task.

Calibration should be measured directly. Score distributions on out-of-distribution inputs (blank images, random noise) are diagnostic of failure modes that mAP cannot detect. We recommend reporting blank-image max score as a basic calibration check for any fine-tuned open-vocabulary detector.

Negative results have evaluative value. Our finding that $9.7\times$ mAP improvement produces zero closed-loop improvement is a result about evaluation methodology, not a result about a particular detector. Such findings deserve publication because they constrain what conclusions can be drawn from offline metrics in this domain.

8 What Yonder Is For (and What It Is Not For)

In light of our findings, we describe the appropriate uses of Yonder.

Appropriate uses:

- Training drone-perspective perception models that are subsequently evaluated either in Habitat-Sim itself (matching training and evaluation distributions) or in a closed-loop benchmark that explicitly accounts for cross-simulator transfer.

- Self-supervised pretraining for representations that will later be fine-tuned on smaller target-domain datasets.
- Studying cross-simulator generalization, including using our fine-tuned checkpoints (which we release) as a reproducible substrate for further experiments.
- Evaluating depth estimation, semantic segmentation, and other tasks where the training/evaluation distribution is held constant.
- Benchmarking detection models within the Habitat-Sim domain.

291 **Inappropriate uses:**

- Training models intended for direct deployment in non-Habitat-Sim environments (Isaac Sim, Unreal Engine, real-world) without target-domain calibration.
- Reporting fine-tuning improvements based solely on Yonder offline metrics. Closed-loop evaluation in the deployment domain is required.
- Training end-to-end navigation policies. Yonder does not contain trajectories with physics, collisions, or replanning.

298 **9 Limitations**

299 Our results are based on indoor environments only; we have not tested cross-simulator transfer for
300 outdoor drone navigation. We test two detector architectures (OWL-ViT v2 and Grounding DINO);
301 other architectures may exhibit different failure modes. We use Habitat-Sim and Isaac Sim as the
302 two simulator endpoints; other simulator pairs may show different gap magnitudes. Real-world
303 transfer (sim-to-real) remains unstudied in this paper.

304 The failure mode diagnosis is based on close inspection of trial logs and 3D goal predictions; alter-
305 native explanations of the closed-loop convergence at 23–25% (e.g., that the binding constraint is
306 actually exploration or planning, not detection) are possible and consistent with our data. We discuss
307 this in companion work [5].

308 We rely on programmatically derived semantic annotations from HSSD’s mesh-authored instance
309 IDs; failure modes inherent to mesh-authored semantics (over-segmentation of articulated objects,
310 occasional misleading boxes around partially-occluded instances) are documented but not corrected.
311 Future work could improve label fidelity using stronger detectors as pseudo-labelers.

312 **10 Conclusion**

313 We introduce Yonder, a 4.65-million-frame drone-perspective dataset for indoor navigation, and use
314 it to investigate whether offline detection improvements predict closed-loop navigation performance.
315 Across four fine-tuning attempts spanning two detector architectures, we find that $9.7\times$ improvement
316 in offline mAP produces zero improvement in closed-loop navigation success—and in some cases,
317 statistically significant degradation. The root cause is a cross-simulator domain gap that manifests as
318 confidence calibration collapse, negative-altitude goal predictions, and lateral localization shifts, but
319 is invisible to standard offline metrics. Detection success rate converges at 23–25% across all con-
320 figurations, fine-tuned and zero-shot, demonstrating that the apparent binding constraint (detection
321 quality) is not the actual constraint.

322 Our findings argue for closed-loop evaluation as a non-negotiable component of any work claim-
323 ing to improve perception for embodied AI tasks, and for explicit testing of cross-simulator trans-
324 fer when training and evaluation use different simulation platforms. We release Yonder, all four
325 fine-tuned model checkpoints, the closed-loop benchmark code, and complete trial logs to enable
326 rigorous study of these phenomena.

327 **References**

- [1] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, C. Li, J. Yang, H. Su, J. Zhu, and L. Zhang. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. *ECCV*, 2024.
- [2] M. Minderer, A. Gritsenko, and N. Houlsby. Scaling open-vocabulary object detection. *NeurIPS*, 2023.

- [3] B. Xiao, H. Wu, W. Xu, X. Dai, H. Hu, Y. Lu, M. Zeng, C. Liu, and L. Yuan. Florence-2: Advancing a unified representation for a variety of vision tasks. *CVPR*, 2024.
- [4] Anonymous. Closing the metric gap: From diagnosis to solution in vision-language drone navigation. 2026. (Anonymized self-citation.)
- [5] Anonymous. Engineering the separation principle: From modular architecture to deployable drone navigation. 2026. (Anonymized self-citation.)
- [6] D. Du, Y. Qi, H. Yu, Y. Yang, K. Duan, G. Li, W. Zhang, Q. Huang, and Q. Tian. The unmanned aerial vehicle benchmark: Object detection and tracking. *ECCV*, 2018.
- [7] H. Huang, F. Zhao, S. Anand, Y. Wang, and L. Zhang. VLMaps: Visual language maps for robot navigation. *IJRR*, 2025.
- [8] D. S. Chaplot, D. Gandhi, S. Gupta, A. Gupta, and R. Salakhutdinov. Learning to explore using active neural SLAM. *ICLR*, 2020.
- [9] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. *IRIS*, 2017.
- [10] F. Sadeghi and S. Levine. CAD2RL: Real single-image flight without a single real image. *RSS*, 2017.
- [11] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, D. Parikh, and D. Batra. Habitat: A platform for embodied AI research. *ICCV*, 2019.
- [12] M. Jacinto, J. Pinto, J. Patrikar, J. Keller, R. Cunha, S. Scherer, and A. Pascoal. Pegasus simulator: An Isaac Sim framework for multiple aerial vehicles simulation. *ICUAS*, 2024.
- [13] P. Anderson, Q. Wu, D. Teney, J. Bruce, M. Johnson, N. Sünderhauf, I. Reid, S. Gould, and A. van den Hengel. Vision-and-language navigation: Interpreting visually-grounded navigation instructions in real environments. *CVPR*, 2018.
- [14] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao. Depth anything V2. *NeurIPS*, 2024.
- [15] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. D. Iii, and K. Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and Section 1 state three claims (dataset scale, offline→closed-loop disconnect, cross-simulator gap diagnosis). Each is supported: dataset scale in Section 3, offline→closed-loop disconnect in Sections 5–6, cross-simulator gap diagnosis in Section 7.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 9 (“Limitations”) discusses indoor-only scope, the two-architecture and two-simulator scope of the empirical claims, the absence of sim-to-real validation, alternative explanations for the 23–25% closed-loop convergence, and limitations inherited from HSSD’s mesh-authored semantic annotations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[N/A\]](#)

Justification: The paper presents an empirical dataset and benchmark with empirical findings; it contains no theoretical results. Equation (1) is a definition of the backprojection used by the navigation pipeline, not a theorem.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3.1 specifies the simulator (Habitat-Sim), drone configuration, way-point sampling rule, and sensor configuration. Section 4 specifies the closed-loop protocol (controller, observation rate, success criterion, timeout, seeds 1000–5000, 93 trials per seed, 465 trials per condition). Section 5.1 specifies fine-tuning hyperparameters (learning rates, frozen layers, label smoothing, sample counts, epoch counts) for each of the four protocols. Dataset, code, and all four fine-tuned checkpoints are released.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The Yonder dataset is openly hosted at <https://huggingface.co/datasets/astralhf/yonder> (CC-BY-NC-4.0); a ~500MB reviewer sample is at <https://huggingface.co/datasets/astralhf/yonder-sample>. All four fine-tuned checkpoints, closed-loop benchmark code, and complete trial logs are released alongside the dataset. The dataset card and Croissant metadata document loading, splits, and preprocessing.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 5.1 lists, per fine-tuning protocol, the architecture, frozen-layer schedule, learning rates (heads vs. backbone), loss function and label smoothing, training-frame counts and number of source scenes, and epoch counts. Section 3 specifies the held-out test split (17 scenes, environment-level not frame-level). Section 4 specifies the closed-loop evaluation protocol.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Section 4.1 describes the statistical protocol: 5 independent random seeds (1000, 2000, 3000, 4000, 5000) per condition, 93 trials per seed, 465 trials per condition; 95% confidence intervals computed via 10,000-resample bootstrap; pairwise significance via 10,000-permutation tests with Bonferroni correction. Tables 2–4 report mean SR@5m with 95% CIs and pairwise p -values against the zero-shot baseline.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Section 3.4 reports the dataset-generation compute (RTX 4090 instances rented from Vast.ai, ~3 hours wall-clock with a fleet of 50 concurrent workers, ~\$60 total). Fine-tuning was performed on a dual RTX 5090 host per protocol; the OWL-ViT v2 fine-tune (29K frames, 5 epochs) and Grounding DINO fine-tunes (8–40K frames) each

428 completed in under 8 wall-clock hours. Closed-loop evaluation totalled $\sim 3,000+$ trials at
429 ≤ 90 s wall-clock per trial in Isaac Sim on a single RTX 4090 host.

430 9. Code of ethics

431 Question: Does the research conducted in the paper conform, in every respect, with the
432 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

433 Answer: [Yes]

434 Justification: The work involves no human subjects, no real persons, no PII, and no biomet-
435 ric data: the dataset is rendered entirely from synthetic 3D scenes. Dual-use risks (surveil-
436 lance, person identification, lethal-autonomy applications) are discussed in the Datasheet
437 (“Uses”) and Section 8, and disallowed uses are encoded both in the CC-BY-NC license
438 and in the explicit usage statement.

439 10. Broader impacts

440 Question: Does the paper discuss both potential positive societal impacts and negative
441 societal impacts of the work performed?

442 Answer: [Yes]

443 Justification: Positive impacts (more rigorous evaluation methodology for embodied AI;
444 reduction of overclaim risk in fine-tuning reports) are discussed in Section 8 and Sec-
445 tion 10. Negative impacts (drone-perspective perception models can in principle be applied
446 to surveillance, person identification, and autonomous-targeting systems) are discussed in
447 Section 8 and the Datasheet. Mitigations: CC-BY-NC license restricts commercial use,
448 including military and surveillance applications, by default; the dataset contains no real
449 persons; explicit disallowed-use list in the Datasheet.

450 11. Safeguards

451 Question: Does the paper describe safeguards that have been put in place for responsible
452 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
453 image generators, or scraped datasets)?

454 Answer: [Yes]

455 Justification: The dataset contains no real persons, no PII, no biometric data, no faces, and
456 no scraped imagery, so the most common scraped-data risks do not apply. The CC-BY-
457 NC-4.0 license forbids commercial use, including commercial military and surveillance
458 applications. The Datasheet enumerates disallowed uses (surveillance, biometric identifica-
459 tion, identification of specific real persons). The released fine-tuned detection checkpoints
460 are domain-specific (Habitat-Sim indoor furniture/objects) and not general-purpose person
461 detectors.

462 12. Licenses for existing assets

463 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
464 the paper, properly credited and are the license and terms of use explicitly mentioned and
465 properly respected?

466 Answer: [Yes]

467 Justification: HSSD (CC-BY-NC-4.0) is cited and credited; Yonder inherits its NonCom-
468 mercial restriction. Habitat-Sim [11] (MIT) and the Pegasus Simulator [12] are cited. The
469 detector backbones used (OWL-ViT v2 [2], Grounding DINO [1]) and Depth Anything
470 V2 [14] are cited with their releases. Replica- and HM3D-derived scenes were excluded
471 because their upstream licenses do not permit open redistribution of derivative renders;
472 ReplicaCAD-derived scenes were excluded because they were not used in any reported
473 experiment.

474 13. New assets

475 Question: Are new assets introduced in the paper well documented and is the documenta-
476 tion provided alongside the assets?

477 Answer: [Yes]

Justification: The Yonder dataset card on the HuggingFace Hub documents repository layout, sensor schema, scene-source provenance, license, intended use, and known limitations. A Croissant metadata file with core and Responsible AI fields accompanies the dataset. A reviewer-friendly ~500 MB sample subset is published as a separate HF repository. The Datasheet for Yonder (Appendix A) follows the structure of [15].

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The work involves no crowdsourcing and no human subjects. All data is rendered from synthetic 3D scenes; semantic annotations are derived programmatically from mesh-authored instance IDs.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human subjects research; IRB approval is not applicable.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not a component of the core methodology. The vision-language detectors evaluated (OWL-ViT v2 and Grounding DINO) are the subject of study, not methodological tools used to perform the analysis; both are cited and described in Section 5. Any LLM use in writing or editing was limited to the kinds of assistance excluded by the policy from declaration.

A Datasheet for Yonder

Following [15]:

Motivation. Created to enable training of drone-perspective perception models for autonomous indoor navigation. Creating organization and funder withheld for double-blind review.

Composition. 4,650,324 frames across 167 indoor 3D scenes (all from HSSD), organized as 387,527 waypoint NPZ files (52 sensor arrays per file plus 12 per-pixel semantic arrays per file, with semantic instance labels for all 167 scenes).

Collection process. Generated by flying a simulated Holybro x500v2 quadcopter through 3D scenes in Habitat-Sim. Waypoint sampling uses the navmesh with adaptive densification. Sensor capture is deterministic given a scene and seed.

Preprocessing. Raw renders compressed to NPZ with gzip level-1. Semantic channels re-rendered separately and projected to COCO bounding boxes by extracting bounding boxes around each unique semantic instance ID with area > 100 pixels.

Uses. Recommended for perception model training with closed-loop validation in deployment domain. Not recommended for end-to-end navigation policy training or for reporting fine-tuning improvements based solely on offline metrics.

Distribution. Hosted on the HuggingFace Hub at <https://huggingface.co/datasets/astralhf/yonder>. Croissant metadata with Responsible AI fields published alongside the dataset

528 card. License: **CC-BY-NC-4.0** (data, inheriting HSSD's NonCommercial restriction; HSSD attribu-
529 tion preserved per source license) and Apache-2.0 (code). Replica and HM3D-derived scenes were
530 excluded because their upstream licenses do not permit open redistribution of derivative renders;
531 ReplicaCAD scenes were excluded because they lack semantic annotations and were not used in
532 any reported experiment.

533 **Maintenance.** Maintained by the corresponding author. Updates and corrections will be posted to
534 the project website.